

COMPSCI5079 (M)

Cryptography & Secure Development

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Lecture 2:

Traditional Encryption and Decryption schemes





Lecture Outline

1. Types of Attack
 - Threats that need to be protected against.
2. Transposition and Substitution ciphers.
3. Homophonic Ciphers
 - Plaintext letters can have several ciphertext letters.
4. Polyalphabet Ciphers and Rotor Machines.
 - Mechanical encryption devices.
5. Running Key Ciphers
 - The key is the same length as the plaintext



Transposition Ciphers



Transposition Ciphers

- Split characters into blocks of fixed length d .
- Rearrange the characters inside a block according to a key-dependent permutation.
- Decryption uses the inverse permutation to recover the original text (i.e., plaintext).

- We can define the permutation by a table, e.g.
 - $0 \rightarrow 1, 1 \rightarrow 3, 2 \rightarrow 0, 3 \rightarrow 2$.
- It is shorter to list the destination positions in order.
 - (1, 3, 0, 2) or 1302
- The inverse permutation is
 - (2, 0, 3, 1) or 2031



Transposition Ciphers: Example

- Starting with the text
 - securityandcryptography
- Arrange it in blocks of 4 characters
 - secu rity andc rypt ogra phy
- Encrypting the text with the above permutation yields
 - csue tryi dacn prty roag yph.
- Note that we have to do something with the short final block of only 3 letters.
 - We redefine the permutation for just 3 letters by making 3 in the ciphertext go to 2.
 - Encryption permutation = 120, decryption = 201.



Unicity Distance

- The first step in calculating the Unicity Distance is to examine the "key-space" in more detail.
- Assume that all permutations are equally likely.
 - Thus, there are $d!$ possible permutations for an encryption with block length d :

$$d! = d \times (d-1) \times (d-2) \times \dots \times 1.$$

d positions for 1st letter, $(d-1)$ for second and so on.

- $d!$ is very large for moderate values of d .



Unicity Distance

- The entropy of the key space is $\log_2(d!)$.
 - Assume all permutations are equally likely.
- We can estimate this using Stirling's approximation for the factorial function:

$$\log_e(n!) \approx (n + 1/2) \log_e(n) - n$$

Now: $\log_2(x) = \log_2(e) \times \log_e(x) = 1.4427 \log_e(x)$

- If $d = 25$ then $H(K) = 83.6767$
 - There are a lot of possible keys.
- Hence $N_u = 83.6767 / 3.2 = 26$
 - 3.2 is the redundancy of English.



Confusion and Diffusion

- This transposition algorithm can result in good confusion.
 - It all depends on the way the notation for the key is related to the permutation.
- Diffusion is poor because the plaintext bits are shuffled round in character-sized blocks.
 - Replacing the first **s** in the plaintext by a **t** does not change the ciphertext very much.



Breaking the Cipher

- Consider the “Ciphertext only attack”.
 - The only information we have is the ciphertext.
- Observation: the letter frequencies in the ciphertext remain the same as those for the plaintext.
 - This is a good indication that a transposition cipher is being used.
- The order of the letters has been changed, and the original order can be recovered by the use of *anagramming techniques*.
 - This is greatly helped by the use of di-gram and tri-gram frequency tables, showing which letter pairs are common, and also which pairs never occur.



Breaking Cipher: Example

- Start by guessing that the permutation length is 4, then look for clues in the blocks.

csue tryi dacn prty roag yph

- Notice that the third block contains the letters for a-n-d, a very common tri-gram.
- If this block contained the word "and", there would be only two possible permutations.
 - c is the first letter in the block "cand" (2130)
 - c is the last letter in the block "andc" (1302).



Breaking Cipher: Example

- Making **c** the first letter in the block leads to
usec yrit cand tryp aogr phy
- It is quite hard to automatically detect that this is wrong.
- Making **c** the last letter of the block leads to the correct plaintext.
- This approach has reduced the number of possible keys that we need to consider.



Quizzes

The plaintext and ciphertext both use an alphabet of 32 characters, the 26 letters of the alphabet and 6 punctuation characters.

- Encryption consists of taking the characters in blocks of 5 and rearranging them to form the ciphertext.
 - The **rearrangement permutation** is the key, and the inverse permutation is used to decrypt.
1. Why is this not a two-key system, since different permutations are used to encrypt and decrypt?
 2. If a **ciphertext-only attack** were used to try and break the encryption, how many letters of ciphertext would be needed?
 3. Suggest a mechanism for dealing with messages that are not an exact multiple of 5 letters long.